PRD: TrustTally / General Store

Version 1.0 · Platforms: Mobile App + Website · App Language: English (Urdu optional) · Status: Draft for Owner Approval

1. Product Summary

The **Digital TrustTally** shopkeeper record customer credit (udhaar), received payments (jama), daily sales, expenses and monthly ledgers in one place. Every customer gets a personal dashboard that shows **only their own account**. The shopkeeper gets a separate dashboard that shows **all of their customers** (maximum 500). The system must keep running reliably for at least **5 years**.

2. Role, Skills and Guardrails (Mandatory)

2.1 Role

Role	Responsibility
Professional Full-Stack Developer (Node.js / Express.js)	Build the frontend, backend and database exactly as this PRD describes
UI/UX Designer	Design and deliver every screen according to Section 9, using the red and white colour shade

Both roles work **only inside this PRD**. Nothing is added, removed or changed without the owner's permission.

2.2 Skills

#	Skill	Used For	Specification
1	**Frontend: HTML**	Page structure for the website and the mobile app	Semantic HTML5, responsive layout
2	**Frontend: CSS**	Styling, red and white theme, blinking alerts, responsive design	CSS3, mobile-first
3	**Frontend: JavaScript**	Screen logic, API calls, offline mode, language switch	Plain JavaScript (no framework unless approved)
4	**Backend: Node.js / Express.js**	REST API, authentication, business rules, scheduled jobs, PDF/Excel export	Node.js with Express.js
5	**UI/UX Design**	Layout, navigation, 8K HD imagery, blinking, red and white shade	Wireframes and mockups for owner approval
6	**Database: SQL** (added last, see Section 16)	Storing all ledger, payment, audit and chat data for 5+ years	SQL relational database

2.3 Guardrails

Scope control

- Build only what is written in this PRD. No extra features, pages, buttons, screens or integrations without the owner's written permission.
- If something is missing or unclear, ask the owner first. Do not guess and do not add on your own.
- Any new idea is reported to the owner as a suggestion at the end of a delivery. It is not built until approved.

Technology

- Frontend: HTML, CSS and JavaScript only. Backend: Node.js with Express.js. Database: SQL.

- No other framework, language or paid service is introduced without approval. Any library needed (for example PDF or Excel export) is listed to the owner for approval first.

Design

- Colour shade is **red and white only**. No other hue is used.
- Blinking is used only where Section 9.3 allows it.
- Status is always shown with a text label and icon, never by colour alone.

Data and security

- A customer can see **only their own data**. The server enforces this on every request.
- The shopkeeper can see **only their own customers**.
- The **audit log is visible to the shopkeeper only**. Customers never see it.
- Passwords, card numbers, CVV and OTPs are never stored in plain text and never shown in chat or reports.

Financial integrity

- Saved bills and payments are never hard-deleted. Corrections are made with reversal, return or cancel entries.
- Every create, update and approve action is written to the audit log.
- An online payment never changes a customer's balance until the **shopkeeper has checked and confirmed it**.
- A shopkeeper cannot have more than **500 customers**.

3. Users and Access Rules

User	Can See	Can Do	Cannot See
Shopkeeper (Owner)	All own customers, bills, payments, reports, expenses, suppliers, staff, audit log	Everything in Section 4	Other shopkeepers' data
Staff (Munshi)	Only what the owner permits	Only the permitted actions	Audit log, and anything not permitted
Customer	Own profile, bills, payments, statements, chat	OK or Dispute items, submit payments, chat	Other customers, audit log, shop expenses and reports

4. Features

4.1 Shopkeeper Features

ID	Feature	Details
S01	Profile	Shopkeeper name, shop name, Shopkeeper ID, contact number, address, photo/logo
S02	Account Details	Bank, Easypaisa and JazzCash account details. Customers see these on the Pay screen
S03	Dashboard	Today's sales, cash received, online received, credit given, total pending, payments awaiting approval, expenses, customers count (x / 500), overdue customers
S04	Customers	Add, edit, deactivate. Name, auto Customer ID, phone, address, opening balance, credit limit. Search and full ledger per customer
S05	Billing	Items with name, quantity (kg / liter / piece), price, line total. Bill total, paid now, pending, date and customer contact number
S06	Payments	Record cash received by hand, approve or reject online payments, payment history. Customer receives an update message
S07	Daily Report	Per customer: items taken, quantity, price, total, paid and remaining. Plus shop totals for the day
S08	Monthly Reports	Per customer statement, previous month paid and unpaid, unpaid bills list, report by payment method

ID	Feature	Details
S09	Expenses	Category, amount, date and note. Daily and monthly expense report
S10	Online Chat	Chat with each customer
S11	Update Messages	Automatic message to the customer on bill, payment, approval, reminder or correction
S12	Audit Log	Create, update and approve actions with who, when, old value and new value (shopkeeper only)

4.2 Customer Features

ID	Feature	Details
C01	Profile	Name, Customer ID, contact number, shop name, shopkeeper name
C02	Dashboard	Total remaining, previous month balance, this month's bills, last payment, payment window banner (5th to 10th)
C03	My Bills	Items taken, quantity, price, amount, paid and remaining. **OK button next to every item**
C04	Pay Screen	Shopkeeper's Bank, Easypaisa and JazzCash details with copy buttons. Submit payment proof for those who cannot visit the shop
C05	My Payments	Payment history with status: Pending, Approved or Rejected
C06	Monthly Statements	Previous month paid and remaining, carried forward into the next month
C07	Shopkeeper Contact	Shop name, shopkeeper name and contact number
C08	Online Chat	Chat with the shopkeeper
C09	Daily and Monthly Report	Own data only

4.3 Approved Additional Features

ID	Feature	Purpose and Behaviour
A01	**Item Catalogue**	Saved items and prices so bills are made faster with fewer mistakes. Price fills in automatically and any change on a bill is logged
A02	**Offline Mode**	Shopkeeper can create bills and record cash payments without internet. Data syncs automatically when the connection returns
A03	**Credit Limit**	A limit per customer so credit does not grow past it. A bill that crosses the limit shows a warning and needs the shopkeeper's confirmation with a reason
A04	**Return / Cancel Item**	Returned items or cancelled bills create a correction entry so the balance stays correct. The customer is notified
A05	**Dispute Button**	Next to OK on every item. The customer can object to a wrong item with a reason. The shopkeeper sees it and resolves it
A06	**PIN / Account Recovery**	Login with phone and PIN. Forgotten PIN or changed phone is recovered through OTP verification
A07	**Backup and Export**	Automatic backups so data is never lost. Export ledgers and reports as PDF and Excel
A08	**Customer Consent**	Before a customer is added to a khata, they receive an invite and must accept it
A09	**WhatsApp / SMS Reminder and Bill Share**	Reminders and bills reach customers who do not use the app. The month's data is sent as a **PDF**

ID	Feature	Purpose and Behaviour
A10	**Staff (Munshi) Account**	Owner gives family members or workers their own login with separate permissions. Their actions appear in the audit log
A11	**Urdu Language**	Language switch between English (default) and Urdu
A12	**Receipt Printing**	Bill and payment receipts for counter use in thermal printer layout
A13	**Supplier Khata**	Records what the shop owes its suppliers: purchases on credit, payments made and balance (shopkeeper only)
A14	**Payment Gateway: Raast / Easypaisa / JazzCash**	Payment is verified automatically with the provider. It is then shown to the shopkeeper, who checks and confirms before it counts
A15	**QR Code Payment**	Customer scans a QR code and pays through Raast, Easypaisa or JazzCash. It follows the same verify, then shopkeeper confirms, rule as A14

5. Business Rules

ID	Rule
BR-1	One shopkeeper can add a maximum of 500 customers
BR-2	Customer balance = Opening balance + all bills − confirmed payments − returns
BR-3	"Paid Now" on a bill is stored as a payment linked to that bill. Pending on the bill = Total − Paid Now
BR-4	A payment is applied to the **oldest unpaid amount first**
BR-5	At month end the statement closes. Closing balance becomes the **Previous Balance** of the next month
BR-6	The payment window is the **5th to the 10th** of each month. Reminders go out on the 5th, 8th and 10th
BR-7	Amounts still unpaid after the 10th are marked **Overdue** and carried forward
BR-8	Online payments stay **Pending** until the shopkeeper confirms them
BR-9	Saved records are never deleted. Corrections use reversal entries
BR-10	Customer sees only own data. Audit log is shopkeeper-only

6. Complete Flows

6.1 Onboarding Flow

1. Shopkeeper registers with phone number and OTP, then sets a PIN. A Shopkeeper ID is created.
2. Shopkeeper completes the profile and adds Bank, Easypaisa and JazzCash account details.
3. Shopkeeper adds items and prices to the Item Catalogue.
4. Shopkeeper adds a customer: name, phone, address, opening balance and credit limit. A Customer ID is created.
5. The customer receives an invite by SMS or WhatsApp and must **accept** it (Customer Consent).
6. After accepting, the customer logs in with phone and OTP, sets a PIN and sees their own dashboard.

6.2 Daily Billing Flow

1. The customer takes goods from the shop.
2. Shopkeeper opens **New Bill** and selects the customer by name or ID.
3. Shopkeeper adds items from the catalogue with quantity (kg / liter / piece). Price and line total fill in automatically.
4. Shopkeeper enters **Paid Now**. The system calculates **Pending**.
5. If the new balance crosses the credit limit, a warning appears and the shopkeeper confirms with a reason.
6. The bill is saved and an entry is written to the audit log. The customer gets an update message.
7. The customer opens the bill and presses **OK** or **Dispute** on each item.
8. The shopkeeper sees every item's status: Pending, OK or Disputed.

6.3 Cash Payment Flow (By Hand)

1. The customer pays cash at the shop.
2. Shopkeeper opens **Receive Payment**, selects the customer, enters the amount and chooses Cash.
3. The payment is saved and applied to the oldest unpaid amount.
4. The customer receives a message with the amount paid and the remaining balance.
5. The audit log records the entry.

6.4 Online Payment Flow (Manual Proof)

1. A customer who cannot visit the shop opens the **Pay** screen and sees the shopkeeper's account numbers.
2. The customer sends the money using their own banking, Easypaisa or JazzCash app.
3. The customer submits a payment proof: amount, method, transaction ID and screenshot.
4. The payment shows **Pending Approval** and the balance does not change yet.
5. The shopkeeper checks their account and presses **Approve** or **Reject** (with a reason).
6. On approval the payment is applied, the balance is reduced and both sides get a message and receipt.

6.5 Gateway and QR Payment Flow (Raast / Easypaisa / JazzCash)

1. The customer chooses **Pay Online** and either scans the QR code or selects Raast, Easypaisa or JazzCash.
2. The customer completes the payment with the provider.
3. The system verifies the transaction with the provider automatically.
4. The payment appears to the shopkeeper as **Verified, Awaiting Confirmation**.
5. The shopkeeper checks it and presses **Confirm**.
6. Only after confirmation is the payment applied and the balance reduced.

6.6 Return / Cancel Flow

1. Shopkeeper opens the bill and selects **Return Item** or **Cancel Bill**, then enters a reason.
2. A correction entry reduces the customer's balance. The original bill stays on record.
3. The customer is notified and the audit log records it.

6.7 Dispute Flow

1. The customer presses **Dispute** on an item and writes a reason.
2. The shopkeeper is notified and the item is marked Disputed.
3. The shopkeeper corrects the item or rejects the dispute with a note. The customer is told the result.

6.8 Monthly Cycle Flow

1. At month end the system closes each customer's statement: Previous Balance + Bills − Payments − Returns = Closing Balance.
2. The Closing Balance appears as the **Previous Balance** in the next month, in both the shopkeeper's and the customer's view.
3. From the 5th to the 10th the customer can pay. Reminders are sent on the 5th, 8th and 10th.
4. Payments made in this window are applied to the oldest unpaid amount first. The paid part is credited and the remaining part stays as balance.
5. After the 10th, unpaid amounts are marked Overdue.
6. The full statement detail is saved for the shopkeeper and the customer.

6.9 Reminder and Bill Share Flow

1. Reminders are sent by WhatsApp or SMS to customers on the due-window dates.
2. The monthly statement is generated as a **PDF** and shared by WhatsApp. SMS carries a secure download link.
3. The shopkeeper can also share any bill the same way.

7. Worked Example

Bill: Ahmed Khan (CUS-0012)

Item	Quantity	Price	Amount	Customer Status
Rice	5 kg	Rs. 280	Rs. 1,400	OK
Cooking Oil	2 liter	Rs. 520	Rs. 1,040	OK
Sugar	3 kg	Rs. 170	Rs. 510	Pending
Bill Total			**Rs. 2,950**	
Paid Now			**Rs. 1,000**	
Pending on this Bill			**Rs. 1,950**	

Monthly Statement: Ahmed Khan

Detail	October	November (so far)
Previous Balance	Rs. 3,000	Rs. 7,500
Bills	+ Rs. 8,500	+ Rs. 3,200
Payments	− Rs. 4,000	− Rs. 5,000 (7 Nov)
Closing / Current Balance	**Rs. 7,500**	**Rs. 5,700**

The Rs. 5,000 paid on 7 November is applied to the October balance first. Rs. 2,500 of the old balance remains and is marked Overdue. Together with Rs. 3,200 of new bills, the current balance is Rs. 5,700.

8. Navigation (Left Sidebar)

On the website the sidebar is always visible on the left. On mobile it opens as a drawer.

Shopkeeper Sidebar	Customer Sidebar
Dashboard	Dashboard
Customers	My Bills
New Bill	Pay
Bills	My Payments
Payments and Approvals	Monthly Statements
Daily Report	Chat
Monthly Reports	Shopkeeper Contact
Expenses	Profile
Suppliers	
Item Catalogue	
Chat	
Account Details	
Staff	

Shopkeeper Sidebar	Customer Sidebar
Audit Log	
Profile and Settings	

9. UI/UX Design Specification

9.1 Colour Shade: Red and White

Token	Shade	Hex	Use
Primary Red	Red	#D32F2F	Buttons, active sidebar item, key figures
Dark Red	Dark red	#B71C1C	Headers, hover and pressed states
Deep Red	Deepest red	#7F0000	Text on white
Light Red	Pale red tint	#FFEBEE	Cards, table stripes, highlights
White	White	#FFFFFF	Page background, text on red

9.2 Imagery: 8K HD Photos

- High-quality **8K HD (7680 x 4320)** photos are used for the login screen, dashboard banners and empty states.
- The 8K originals are kept as masters. The app serves optimised responsive versions so screens still load within the performance targets.

9.3 Blinking

Blinking draws attention to important states only: **Overdue balance**, **payment waiting for approval**, **unread chat message**, **disputed item** and the **5th to 10th payment window banner**.

- Blink speed is slow (about once per second) and stops once the user opens the item.
- Blinking is switched off for users who have reduced-motion turned on in their device.

9.4 Layout and Behaviour

- Same design language on website and mobile app. Left sidebar on website, drawer on mobile.
- Large touch-friendly buttons. The bill screen is designed for fast use at the counter.
- Status labels (OK, Pending, Disputed, Approved, Rejected, Overdue) use text and icon, not colour alone.
- Automatic Item Picture Reference: When the shopkeeper adds an item to a bill, the system will automatically send or display the relevant item picture to the customer alongside the item name and billing details. This will help the customer easily identify and verify the billed item. If no suitable picture is available, the item name and details will still be displayed normally. The shopkeeper can also add or update an item's picture when needed.

9.5 Screens

Shopkeeper: Login, Dashboard, Customers, Customer Ledger, New Bill, Bills, Payments and Approvals, Daily Report, Monthly Reports, Expenses, Suppliers, Item Catalogue, Chat, Account Details, Staff, Audit Log, Profile.

Customer: Login, Dashboard, My Bills, Pay, My Payments, Monthly Statements, Chat, Shopkeeper Contact, Profile.

10. Technical Architecture

Layer	Specification
Frontend	HTML, CSS and JavaScript. Responsive. Mobile app delivered as an installable app built from the same code (see Section 15)
Offline	Browser service worker and local storage so bills and cash payments work offline, then sync
Backend	Node.js with Express.js REST API
Authentication	Phone + OTP, PIN login, token-based sessions, role checks for Shopkeeper, Staff and Customer
Data isolation	Every query is filtered by the shop and customer taken from the login token, never from user input
Scheduled jobs	Month-end statements, reminders on the 5th, 8th and 10th, daily backup
Exports	PDF and Excel for statements, reports and ledgers
Integrations	WhatsApp/SMS sending and Raast, Easypaisa and JazzCash payment verification

API modules: auth, shops, customers, items, bills, returns, disputes, payments, statements, reports, expenses, suppliers, staff, chat, notifications, audit, export.

11. Non-Functional Requirements

Area	Requirement
5-year operation	Roughly 500 customers x 1 bill a day x 365 x 5 years is about 900,000 bills. The database must handle this without slowing down
Performance	Screens load in 3 seconds or less. A bill saves in 2 seconds or less
Backup	Automatic daily backup with restore testing
Retention	All records kept for at least 5 years
Security	HTTPS, hashed PINs, encrypted sensitive data, rate-limited login, role-based access and session base authentication and add try catch function for role back function in payment method
Compatibility	Latest Android and iOS devices, and modern desktop browsers
Accessibility	Readable contrast, reduced-motion support, large touch targets

12. Benefits and Drawbacks

Benefits	Drawbacks	Mitigation
Records never get lost like a paper khata	Data entry takes time at first	Item catalogue and a quick bill screen
Customers see a clear account, so fewer disputes	Some customers have no smartphone or internet	WhatsApp/SMS reminders and PDF statements
Faster recovery of credit with automatic reminders	App needs internet	Offline mode with automatic sync
Customers can pay without visiting the shop	Online payments need manual checking	Verify first, then shopkeeper confirms
Audit log protects the shopkeeper's records	Shopkeeper carries data privacy responsibility	Encryption, access control, backups
Reports for daily, monthly and yearly review	Ongoing hosting and messaging cost	Phased launch

13. Release Phases

Phase	Scope
Phase 1: Core	Login, PIN and recovery, profiles, customers with consent, item catalogue, billing, cash payments, ledger, customer dashboard, audit log
Phase 2: Payments and Communication	Online payment proof and approval, chat, update messages, monthly cycle and statements, WhatsApp/SMS reminders and PDF sharing
Phase 3: Control and Reports	Daily and monthly reports, expenses, credit limit, returns, disputes, backup and export, staff accounts, Urdu, receipt printing, supplier khata
Phase 4: Gateway and Launch	Raast, Easypaisa, JazzCash and QR payments, full testing, pilot with one shop, launch

14. Acceptance Criteria

1. A shopkeeper can add up to 500 customers and the 501st is blocked.
2. A customer can log in and sees only their own data.
3. A bill shows items, quantity in kg / liter, price, total, paid and pending.
4. The customer can press OK or Dispute on every item.
5. A cash payment immediately reduces the customer's balance and sends a message.
6. An online or QR payment does not change the balance until the shopkeeper confirms it.
7. Closing balance of a month appears as Previous Balance in the next month for both users.
8. Payments made from the 5th to the 10th are applied to the oldest balance first.
9. The audit log records create, update and approve actions and is visible only to the shopkeeper.
10. Bills and payments can be created offline and sync without duplicates.
11. Monthly statement PDF can be shared by WhatsApp and downloaded through an SMS link.
12. The whole app uses only red and white shades, and blinking follows Section 9.3.

15. Assumptions and Open Questions

1. **Item OK button:** assumed the customer presses OK on each item and the shopkeeper sees the status. Please confirm.
2. **Mobile app:** built from the same HTML, CSS and JavaScript as an installable app. A separate store-published native app is not included unless approved.
3. **SQL database engine** (MySQL).
4. **8K photos:** the owner supplies them, or approves licensed images.
5. **WhatsApp/SMS provider and Raast / Easypaisa / JazzCash merchant access** are needed from the owner. Until they are ready, the manual proof flow in 6.4 is used.
6. **Changed phone number:** the exact verification steps for customers and shopkeepers are to be agreed.
7. **Thermal printing:** direct printer connection depends on the device. Browser print is used until confirmed.
8. **Multiple shopkeepers:** assumed that many shopkeepers can use the system, each with up to 500 customers and fully separate data.
9. **Currency:** Pakistani Rupees (Rs.).

16. Database (SQL)

All money values use fixed-point decimals (for example DECIMAL(12,2)). Every shop-owned table carries a shop reference so data can always be separated per shopkeeper.

Table	Key Columns
shops	shop_id, owner_name, shop_name, phone, address, logo
users	user_id, role (shopkeeper / staff / customer), shop_id, phone, pin_hash, language, status
staff_permissions	user_id, permission
customers	customer_id, shop_id, user_id, customer_code (CUS-0001), name, phone, address, opening_balance, credit_limit, consent_status, status

Table	Key Columns
items	item_id, shop_id, name, unit (kg / liter / piece), default_price, active
bills	bill_id, shop_id, customer_id, bill_no, bill_date, total, status (active / cancelled), created_by
bill_items	bill_item_id, bill_id, item_id, name, quantity, unit, price, line_total, customer_status (pending / ok / disputed)
disputes	dispute_id, bill_item_id, reason, status, resolution_note
returns	return_id, bill_id, bill_item_id, quantity, amount, reason, created_by
payments	payment_id, shop_id, customer_id, bill_id (optional), amount, method (cash / bank / easypaisa / jazzcash / raast / qr), transaction_ref, proof_file, status (pending / verified / approved / rejected), confirmed_by, confirmed_at
payment_allocations	payment_id, bill_id, amount
monthly_statements	statement_id, customer_id, month, previous_balance, bills_total, payments_total, returns_total, closing_balance, pdf_path
expenses	expense_id, shop_id, expense_date, category, amount, note
suppliers	supplier_id, shop_id, name, phone
supplier_transactions	transaction_id, supplier_id, type (purchase / payment), amount, date, note
chat_messages	message_id, shop_id, customer_id, sender_id, text, sent_at, read_at
notifications	notification_id, user_id, channel (in-app / whatsapp / sms), type, message, status, sent_at
audit_logs	log_id, shop_id, user_id, action, entity, entity_id, old_value, new_value, reason, created_at

Database rules

- Customer code is unique within a shop. The 500-customer limit is enforced by the backend and checked in the database.
- No hard deletes on bills, payments or returns. Corrections are new rows.
- `audit_logs` is append-only: the application account can insert and read, but cannot update or delete.
- Indexes on shop, customer and date columns keep reports fast through 5+ years of data.
- Balance is calculated from bills, confirmed payments and returns, and matches the monthly statements.
- Daily backups and a retention period of at least 5 years.